

Distributed Fixed-Time Secondary Frequency Control for Incremental-Cost-Droop based Islanded Microgrids

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Abstract: This paper presents a distributed secondary frequency restoration scheme for incremental cost droop-controlled microgrids. The proposed scheme achieves accurate frequency restoration and optimal active power sharing within the fixed time. The upper bound of settling time is independent of initial conditions, which makes it possible to predesign the settling time of the control system. The fixed-time convergence is rigorously proved by applying Lyapunov theory. Finally, a case study is conducted to demonstrate the effectiveness of the proposed scheme.

Key Words: Islanded Microgrids, Distributed Control, Frequency Restoration, Optimal Active Power Sharing, Fixed-Time

1 Introduction

Microgrid is a promising solution for making full use of distributed generation(DG), which can operate in either grid-connected mode or islanded mode. The islanded microgrid not only needs to achieve frequency synchronization and voltage stability, but also needs to achieve active and reactive power sharing. Droop control method is widely used to coordinate distributed generators in microgrid [1-4]. However, droop control leads to the frequency/voltage deviation and the non-optimal power sharing. In order to overcome these problems, hierarchical control strategy(namely the primary droop control, secondary frequency and voltage restoration, and tertiary power optimization) is introduced into microgrid [5-6]. The conventional secondary and tertiary control schemes are implemented in a centralized architecture based on a central controller of microgrid, which collects the global information and provides control command. This method suffers from single point-of-failures, high communication and computation burden, and poor scalability.

In recent years, distributed control schemes for microgrid have attracted a great deal of attention due to the less information requirement, the stronger scalability and reliability [7-8]. In [9], secondary frequency control and active power sharing are converted to the leader-follower consensus problem and the leaderless consensus problem of multi-agent system by feedback linearization. The work in [10] proposed a secondary frequency and voltage control scheme based on distributed-averaging proportional-integral. A distributed secondary control scheme with unreliable communication network is presented in [11]. The work in [12] presents a distributed secondary controller based on internal model principle. In [9-12], the secondary control is achieved with a asymptotical rate, i.e., infinite-time horizon. In order to achieve high-speed convergence, several finite-time secondary control schemes have been proposed. In [13], a binary protocol based distributed finite-time frequency controller is proposed to recover the frequency and

share the active power. A distributed finite-time secondary frequency and voltage control scheme is proposed in [14]. The work in [15] presented a distributed finite-time frequency restoration and active power sharing scheme with bounded control input. The work in [16] proposed a distributed robust finite-time secondary frequency and voltage control scheme, which considers unmodeled dynamics, unknown disturbances and uncertainties. In [17], distributed finite-time secondary control under switching communication topology is designed.

The finite-time controls have advantages of faster convergence, higher precision, robustness against uncertainties and better disturbance rejection, but one shortcoming is that the convergence time of finite-time schemes is closely dependent upon initial states. It poses a challenge to the off-line predesign of the convergence time since the initial states are unavailable in advance. To overcome this shortcoming, fixed-time stability and fixed-time consensus are proposed in [18-21] such that the convergence time is independent of the initial states. In [22], a distributed fixed-time secondary frequency and voltage control scheme is proposed. The work in [23] presented a fixed-time distributed secondary voltage controller. However, the works in [9-17], [22] only achieve the proportional active power sharing, i.e., the total generation cost is not the minimum.

In order to minimize the total generation cost of microgrids, several optimal active power control schemes have been proposed. In [24], an equal increment cost principle based droop control strategy is proposed to minimize the total generation cost. However, the frequency restoration is not considered. The work in [25] presented a distributed optimal active power sharing control scheme based on the hierarchical control. The work in [26] directly optimizes the parameter of droop controller based on the distributed economic dispatch solution, which achieves frequency restoration and optimal active power sharing simultaneously.

Motivated by increment cost droop and fixed-time theory, this paper presents a distributed fixed-time secondary control scheme. The main contributions of the presented controller are twofold. Firstly, frequency restoration and optimal active power sharing are achieved within fixed-time regardless of the initial states, which means that the convergence time can be off-line predesigned. Secondly, convergence property of

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the proposed controller is rigorously proved by fixed-time Lyapunov theory. Moreover, numerical simulations show the effectiveness of the proposed scheme.

The rest of this paper is organized as follows. In section 2, the problem formulation and preliminaries are given. In Section 3, distributed fixed-time secondary frequency controller is designed and analyzed. A case study is given to show the effectiveness of the proposed controller in section 4. Section 5 concludes this paper.

Notations: Let $1_n = [1, 1, \dots, 1]^T \in R^n$, $0_n = [0, 0, \dots, 0]^T \in R^n$, and $\text{span}(1_n) = \{\xi \in R^n : \xi = h1_n, h \in R\}$. For real vector $s = [s_1, s_2, \dots, s_n]^T$, define $s^{[c]} = [s_1^{[c]}, s_2^{[c]}, \dots, s_n^{[c]}]^T$, where $s_i^{[c]} = \text{sign}(s_i)|s_i|^c$, $\text{sign}(\cdot)$ denotes the sign function and $|\cdot|$ denotes the absolute value.

2 Problem Formulation and Preliminaries

2.1 Problem Formulation

Fig.1 shows the distributed control structure of microgrids, which is composed of physical network and cyber network. The physical network mainly contains inverter-based DGs, local controllers, local loads, and electrical network. The cyber network mainly consists of transceivers and communication links. Each inverter-based DG in microgrid consists of a DC energy source, a three-leg voltage-controlled source (VCS) inverter, a LC filter, and a RL output connector. The local controller consists of primary control(i.e., power controller, voltage controller, and current controller) and secondary control(i.e., frequency restoration controller). The distributed control diagram of a single DG is shown in Fig.2.

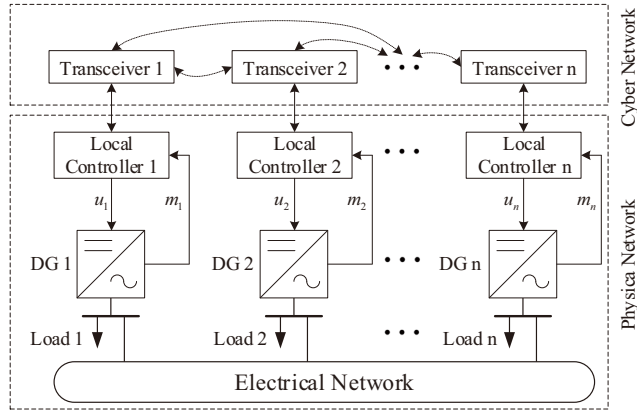


Fig. 1: The distributed control structure of microgrids

Droop control is widely used to maintain the frequency/voltage stability and keep the active/reactive sharing. This paper considers the increment cost principle based droop control strategy [24]

$$\omega_i = \omega_i^* - k\eta_i(P_i) \quad (1)$$

$$v_i = V_i^* - m_{qi}Q_i \quad (2)$$

where ω_i and v_i are output frequency and voltage amplitude of the i th DG, respectively; ω_i^* and V_i^* respectively denote primary frequency and voltage amplitude reference; k and

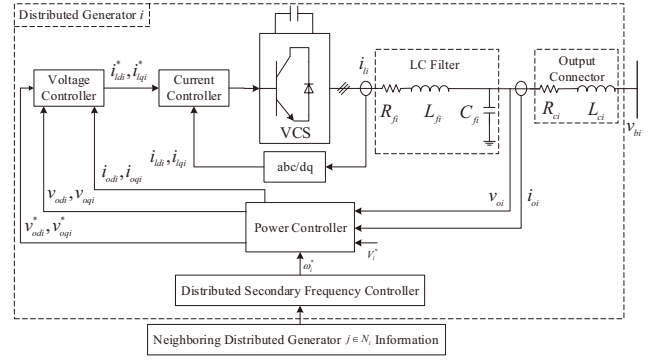


Fig. 2: The distributed control diagram of single DG

m_{qi} are the droop coefficients of frequency and voltage amplitude; P_i and Q_i are active and reactive power outputs of the i th DG, respectively; $\eta_i(P_i)$ is the incremental cost of the i th DG, which is the derivative of the convex cost function $C_i(P_i)$.

The main objectives of this paper are twofold: (1) Restore the output frequency ω_i to the nominal frequency ω^* ; (2) Achieve the optimal active power sharing while maintaining the supply-demand balance, i.e., the incremental cost of each DG is equal, $\eta_1(P_1) = \dots = \eta_n(P_n)$ [25-26].

2.2 Preliminaries

The communication network of microgrids can be represented as an undirected graph, where DGs and communication links are denoted as graph vertices and graph edges, respectively. The undirected graph is usually expressed as $G = (V, E, A)$, where $V = \{1, 2, \dots, n\}$ is a non-empty finite set of vertices, $E \subseteq V \times V$ is a set of edges, and $A = [a_{ij}] \in R^{n \times n}$ is the weighted adjacency matrix. If the vertices i and j exchange information through an undirected edge $(i, j) \in E$, then $a_{ij} = a_{ji} > 0$; otherwise, $a_{ij} = a_{ji} = 0$. The set of neighbours of node i is denoted as $N_i = \{j \in V | (i, j) \in E\}$. Define $L = [l_{ij}] \in R^{n \times n}$ as the Laplacian matrix with $l_{ii} = \sum_{j \in N_i} a_{ij}$ and $l_{ij} = -a_{ij}, i \neq j$. An undirected path between vertex i and vertex j is a sequence of edges, expressed as $\{(i, i_1), (i_1, i_2), \dots, (i_m, j)\}$. An undirected graph G is connected if there exists a path between any distinct pair of vertices. Define $H = \text{diag}\{g_1, g_2, \dots, g_n\}$ as the pinning matrix with pinning gain g_i , where $g_i > 0$ if the i th DG has access to the nominal values; otherwise, $g_i = 0$.

For an undirected and connected graph, if $1_n^T x = 0$, $x^T L x \geq \lambda_2(L) x^T x$, where $\lambda_2(L) > 0$ is the second smallest eigenvalue of L . Moreover, if $H \neq 0$, $x^T (L + H) x \geq \lambda_1(L + H) x^T x$, where $\lambda_1(L + H) > 0$ is the smallest eigenvalue of $L + H$.

Lemma 1: [18] Consider the system $\dot{x} = g(t, x)$, $x(0) = x_0$ with $g(t, 0) = 0$. If there exists a radially unbounded continuous function $V : R^n \rightarrow R_+ \cup \{0\}$ such that 1) $V(x) = 0 \Leftrightarrow x = 0$; 2) any solution $x(t)$ of the system satisfies the inequality $D^+ V(x(t)) \leq -(\alpha V^p(x(t)) + \beta V^q(x(t)))^k$ for some $\alpha, \beta, p, q, k > 0 : pk < 1, qk > 1$, where $D^+ V(x(t))$ denotes the upper right-hand derivative of function $V(x(t))$. Then the origin is globally fixed-time stable, and the follow-

ing estimate of the settling time holds

$$T(x_0) \leq 1/[\alpha^k(1-pk)] + 1/[\beta^k(qk-1)], \forall x_0 \in R^n \quad (3)$$

Lemma 2: [21] Let $\zeta_1, \zeta_2, \dots, \zeta_n \geq 0$, $0 < \phi \leq 1$, $\varphi > 1$. Then

$$\sum_{i=1}^n \zeta_i^\phi \geq (\sum_{i=1}^n \zeta_i)^\phi, \sum_{i=1}^n \zeta_i^\varphi \geq n^{1-\varphi} (\sum_{i=1}^n \zeta_i)^\varphi \quad (4)$$

3 Distributed Fixed-Time Frequency Control

In order to achieve frequency restoration and active power sharing, differentiating the equation (1) yields

$$\dot{\omega}_i = \dot{\omega}_i^* - k\dot{\eta}_i(P_i) \quad (5)$$

Then, (5) can be written as

$$\dot{\omega}_i^* = \dot{\omega}_i + k\dot{\eta}_i(P_i) = u_{\omega i} + u_{\eta i} \quad (6)$$

where $u_{\omega i} = \dot{\omega}_i$ and $u_{\eta i} = k\dot{\eta}_i(P_i)$ are frequency auxiliary control input and incremental cost auxiliary control input, respectively.

Based on each DG's own information and the information of its neighbors on the communication graph, the auxiliary control inputs $u_{\omega i}$ and $u_{\eta i}$ are designed as follows

$$u_{\omega i} = c_\omega \left[\sum_{j \in N_i} a_{ij} (\omega_j - \omega_i)^{[\mu]} + g_i (\omega^* - \omega_i)^{[\mu]} \right] + c_\omega \left[\sum_{j \in N_i} a_{ij} (\omega_j - \omega_i)^{[\nu]} + g_i (\omega^* - \omega_i)^{[\nu]} \right] \quad (7)$$

$$u_{\eta i} = c_\eta \left[\sum_{j \in N_i} a_{ij} (k\eta_j(P_j) - k\eta_i(P_i))^{[\mu]} \right] + c_\eta \left[\sum_{j \in N_i} a_{ij} (k\eta_j(P_j) - k\eta_i(P_i))^{[\nu]} \right] \quad (8)$$

where $c_\omega > 0$ and $c_\eta > 0$ are control gains, $0 < \mu < 1$ and $\nu > 1$ are the control parameters. According to (6), (7), and (8), the primary frequency reference ω_i^* can be written as follows

$$\omega_i^* = \int (u_{\omega i} + u_{\eta i}) dt \quad (9)$$

Next, we give our main results.

Theorem 1: Suppose that the communication graph is connected and at least one DG has access to the nominal frequency value. By using the control protocol (9), the frequency restoration and optimal active power sharing are achieved in fixed-time.

Proof: Define the frequency error term as $e_{\omega i} = \omega_i - \omega^*$. Differentiating $e_{\omega i}$ yields

$$\begin{aligned} \dot{e}_{\omega i} &= \dot{\omega}_i - \dot{\omega}^* = \dot{\omega}_i = u_{\omega i} \\ &= c_\omega \left[\sum_{j \in N_i} a_{ij} (e_{\omega j} - e_{\omega i})^{[\mu]} - g_i e_{\omega i}^{[\mu]} \right] \\ &\quad + c_\omega \left[\sum_{j \in N_i} a_{ij} (e_{\omega j} - e_{\omega i})^{[\nu]} - g_i e_{\omega i}^{[\nu]} \right] \end{aligned} \quad (10)$$

Define the incremental cost error term as $e_{\eta i} = k\eta_i(P_i) - \frac{1}{n} \sum_{i=1}^n k\eta_i(P_i)$. Since the communication graph is undirected, we have $\frac{1}{n} \sum_{i=1}^n k\eta_i(P_i) = \frac{1}{n} \sum_{i=1}^n u_{\eta i} = 0$. Therefore, the term $\frac{1}{n} \sum_{i=1}^n k\eta_i(P_i)$ is time invariant. Differentiating $e_{\eta i}$ yields

$$\begin{aligned} \dot{e}_{\eta i} &= k\dot{\eta}_i(P_i) - \frac{1}{n} \sum_{i=1}^n k\dot{\eta}_i(P_i) = k\dot{\eta}_i(P_i) = u_{\eta i} \\ &= c_\eta \left[\sum_{j \in N_i} a_{ij} (e_{\eta j} - e_{\eta i})^{[\mu]} \right] \\ &\quad + c_\eta \left[\sum_{j \in N_i} a_{ij} (e_{\eta j} - e_{\eta i})^{[\nu]} \right] \end{aligned} \quad (11)$$

Choose the following Lyapunov function

$$\begin{aligned} V_1 &= V_\omega(e_\omega) + V_\eta(e_\eta) = \frac{1}{2} e_\omega^T e_\omega + \frac{1}{2} e_\eta^T e_\eta \\ &= \frac{1}{2} \sum_{i=1}^n e_{\omega i}^2 + \frac{1}{2} \sum_{i=1}^n e_{\eta i}^2 \end{aligned} \quad (12)$$

where $e_\omega = [e_{\omega 1}, \dots, e_{\omega n}]^T$, $e_\eta = [e_{\eta 1}, \dots, e_{\eta n}]^T$.

Differentiating (12) yields

$$\begin{aligned} \dot{V}_1 &= \sum_{i=1}^n e_{\omega i} \dot{e}_{\omega i} + \sum_{i=1}^n e_{\eta i} \dot{e}_{\eta i} \\ &= \sum_{i=1}^n c_\omega e_{\omega i} \left[\sum_{j \in N_i} a_{ij} (e_{\omega j} - e_{\omega i})^{[\mu]} - g_i e_{\omega i}^{[\mu]} \right] \\ &\quad + \sum_{i=1}^n c_\omega e_{\omega i} \left[\sum_{j \in N_i} a_{ij} (e_{\omega j} - e_{\omega i})^{[\nu]} - g_i e_{\omega i}^{[\nu]} \right] \\ &\quad + \sum_{i=1}^n c_\eta a_{ij} e_{\eta i} [(e_{\eta j} - e_{\eta i})^{[\mu]} + (e_{\eta j} - e_{\eta i})^{[\nu]}] \\ &= -\frac{c_\omega}{2} \sum_{i,j=1}^n a_{ij} |e_{\omega j} - e_{\omega i}|^{1+\mu} - c_\omega \sum_{i=1}^n g_i |e_{\omega i}|^{1+\mu} \\ &\quad - \frac{c_\omega}{2} \sum_{i,j=1}^n a_{ij} |e_{\omega j} - e_{\omega i}|^{1+\nu} - c_\omega \sum_{i=1}^n g_i |e_{\omega i}|^{1+\nu} \\ &\quad - \frac{c_\eta}{2} \sum_{i,j=1}^n a_{ij} |e_{\eta j} - e_{\eta i}|^{1+\mu} \\ &\quad - \frac{c_\eta}{2} \sum_{i,j=1}^n a_{ij} |e_{\eta j} - e_{\eta i}|^{1+\nu} \end{aligned} \quad (13)$$

Define $\Gamma(e_\omega^\mu) = \frac{c_\omega}{2} \sum_{i,j=1}^n a_{ij} |e_{\omega j} - e_{\omega i}|^{1+\mu} + c_\omega \sum_{i=1}^n g_i |e_{\omega i}|^{1+\mu}$. According to Lemma 2, one has

$$\begin{aligned} \Gamma(e_\omega^\mu) &\geq \frac{c_\omega}{2} \left(\sum_{i,j=1}^n a_{ij}^{\frac{2}{1+\mu}} (e_{\omega j} - e_{\omega i})^2 \right)^{\frac{1+\mu}{2}} \\ &\quad + 2 \sum_{i=1}^n g_i^{\frac{2}{1+\mu}} (e_{\omega i})^2)^{\frac{1+\mu}{2}} \\ &= \frac{c_\omega}{2} (2e_\omega^T (L^\mu + H^\mu) e_\omega)^{\frac{1+\mu}{2}} \\ &\geq \frac{c_\omega}{2} (2\lambda_1 (L^\mu + H^\mu) e_\omega^T e_\omega)^{\frac{1+\mu}{2}} \\ &= \frac{c_\omega}{2} (4\lambda_1 (L^\mu + H^\mu) V_\omega(e_\omega))^{\frac{1+\mu}{2}} \end{aligned} \quad (14)$$

where L^μ denotes the Laplacian matrix, whose corresponding adjacency matrix is $A^\mu = [a_{ij}^{\frac{2}{1+\mu}}]$; H^μ denotes the pinning matrix with the i th diagonal element $g_i^{\frac{2}{1+\mu}}$.

Similarly, define $\Gamma(e_\omega) = \frac{c_\omega}{2} \sum_{i,j=1}^n a_{ij} |e_{\omega j} - e_{\omega i}|^{1+\nu} + c_\omega \sum_{i=1}^n b_i |e_{\omega i}|^{1+\nu}$, we have

$$\Gamma(e_\omega) \geq \frac{c_\omega}{2} n^{1-\nu} (4\lambda_1(L^\nu + H^\nu) V_\omega(e_\omega))^{\frac{1+\nu}{2}} \quad (15)$$

where L^ν denotes the Laplacian matrix with the corresponding adjacency matrix $A^\nu = [a_{ij}^{\frac{2}{1+\nu}}]$; H^ν denotes the pinning matrix with the i th diagonal element $\frac{1}{2}(2g_i)^{\frac{2}{1+\nu}}$.

Define $\Gamma(e_\eta) = \frac{c_\eta}{2} \sum_{i,j=1}^n a_{ij} |e_{\eta j} - e_{\eta i}|^{1+\mu}$. According to Lemma 2, one has

$$\begin{aligned} \Gamma(e_\eta) &\geq \frac{c_\eta}{2} \left(\sum_{i,j=1}^n a_{ij}^{\frac{2}{1+\mu}} (e_{\eta j} - e_{\eta i})^2 \right)^{\frac{1+\mu}{2}} \\ &= \frac{c_\eta}{2} (2e_\eta^T L^\mu e_\eta)^{\frac{1+\mu}{2}} \geq \frac{c_\eta}{2} (2\lambda_2(L^\mu) e_\eta^T e_\eta)^{\frac{1+\mu}{2}} \\ &= \frac{c_\eta}{2} (4\lambda_2(L^\mu) V_\eta(e_\eta))^{\frac{1+\mu}{2}} \end{aligned} \quad (16)$$

Similarly, by defining $\Gamma(e_\eta) = \frac{c_\eta}{2} \sum_{i,j=1}^n a_{ij} |e_{\eta j} - e_{\eta i}|^{1+\nu}$, we have

$$\Gamma(e_\eta) \geq \frac{c_\eta}{2} n^{1-\nu} (4\lambda_2(L^\nu) V_\eta(e_\eta))^{\frac{1+\nu}{2}} \quad (17)$$

Combining (13), (14), (15), (16), and (17), one has

$$\begin{aligned} \dot{V}_1 &\leq -\frac{c_\omega}{2} (4\lambda_1(L^\mu + H^\mu) V_\omega(e_\omega))^{\frac{1+\mu}{2}} \\ &\quad - \frac{c_\omega}{2} n^{1-\nu} (4\lambda_1(L^\nu + H^\nu) V_\omega(e_\omega))^{\frac{1+\nu}{2}} \\ &\quad - \frac{c_\eta}{2} (4\lambda_2(L^\mu) V_\eta(e_\eta))^{\frac{1+\mu}{2}} \\ &\quad - \frac{c_\eta}{2} n^{1-\nu} (4\lambda_2(L^\nu) V_\eta(e_\eta))^{\frac{1+\nu}{2}} \\ &= -\frac{c_\omega}{2} (4\lambda_1(L^\mu + H^\mu))^{\frac{1+\mu}{2}} (V_\omega(e_\omega))^{\frac{1+\mu}{2}} \\ &\quad - \frac{c_\omega}{2} n^{1-\nu} (4\lambda_1(L^\nu + H^\nu))^{\frac{1+\nu}{2}} (V_\omega(e_\omega))^{\frac{1+\nu}{2}} \\ &\quad - \frac{c_\eta}{2} (4\lambda_2(L^\mu))^{\frac{1+\mu}{2}} (V_\eta(e_\eta))^{\frac{1+\mu}{2}} \\ &\quad - \frac{c_\eta}{2} n^{1-\nu} (4\lambda_2(L^\nu))^{\frac{1+\nu}{2}} (V_\eta(e_\eta))^{\frac{1+\nu}{2}} \end{aligned} \quad (18)$$

Let $\chi = \min\{\frac{c_\eta}{2} (4\lambda_2(L^\mu))^{\frac{1+\mu}{2}}, \frac{c_\omega}{2} (4\lambda_1(L^\mu + H^\mu))^{\frac{1+\mu}{2}}\}$, $\psi = \min\{\frac{c_\omega}{2} n^{1-\nu} (4\lambda_1(L^\nu + H^\nu))^{\frac{1+\nu}{2}}, \frac{c_\eta}{2} n^{1-\nu} (4\lambda_2(L^\nu))^{\frac{1+\nu}{2}}\}$. Then we have

$$\begin{aligned} \dot{V}_1 &\leq -\chi \left((V_\omega(e_\omega))^{\frac{1+\mu}{2}} + (V_\eta(e_\eta))^{\frac{1+\mu}{2}} \right) \\ &\quad - \psi \left((V_\omega(e_\omega))^{\frac{1+\nu}{2}} + (V_\eta(e_\eta))^{\frac{1+\nu}{2}} \right) \\ &\leq -\chi (V_1)^{\frac{1+\mu}{2}} - \psi 2^{\frac{1-\nu}{2}} (V_1)^{\frac{1+\nu}{2}} \end{aligned} \quad (19)$$

Therefore, $\dot{V}_1 \leq -(\alpha V_1^p + \beta V_1^q)^k$, where $k = 1$, $\alpha = \chi$, $\beta = \psi 2^{\frac{1-\nu}{2}}$, $0 < p = \frac{1+\mu}{2} < 1$, $q = \frac{1+\nu}{2} > 1$. Applying Lemma 1, we get that $V_1 \rightarrow 0$ in fixed time with the settling time $T_1 \leq \frac{2}{\chi(1-\mu)} + \frac{2}{\psi 2^{\frac{1-\nu}{2}}(\nu-1)}$. $V_1 = 0$ means that $e_\omega = 0_n$ and $e_\eta = 0_n$, i.e., $\omega_i = \omega^*$ and $\eta_i(P_i) = \eta_j(P_j)$, $\forall i, j$. Thus the frequency restoration and the optimal active power sharing are achieved in fixed time. This completes the proof of Theorem 1.

4 Simulation Study

The AC (50Hz, 311V) islanded microgrid is used to verify the effectiveness of the proposed fixed-time frequency controller. Fig.3(a) depicts the block diagram of the islanded microgrid test system, the corresponding parameters of the microgrid system are adopted from [3,9], which are listed in Table 1. In this simulation, the cost function of DGs is quadratic function [25-26], i.e., $C_i(P_i) = \gamma_i P_i^2 + \rho_i P_i + \delta_i$, the cost coefficients are listed in Table 2. Fig 3 (b) shows the communication topology of microgrid, where only DG 1 has access to the nominal frequency value. In this simulation, the control parameters in (7) and (8) are selected as $c_\omega = c_\eta = 30$, $\mu = 0.5$, $\nu = 2$.

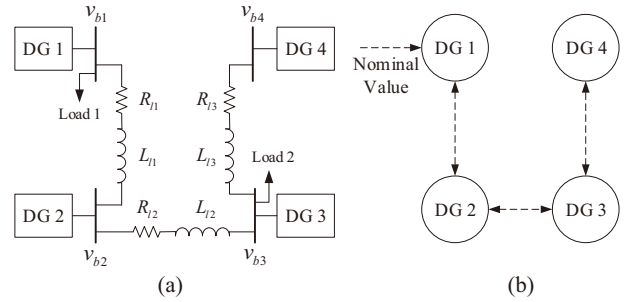


Fig. 3: (a) the block diagram of the microgrid test system, (b) the communication topology of the microgrid

Table 1: Parameters of The Microgrid System

DG i	DG 1	DG 2		DG 3	DG 4
k	6×10^{-4}	6×10^{-4}		6×10^{-4}	6×10^{-4}
m_{qi}	9×10^{-4}	7×10^{-4}		9×10^{-4}	7×10^{-4}
R_f	0.1Ω	0.1Ω		0.1Ω	0.1Ω
L_f	$1.35mH$	$1.35mH$		$1.35mH$	$1.35mH$
C_f	$50\mu F$	$50\mu F$		$50\mu F$	$50\mu F$
R_c	0.03Ω	0.03Ω		0.03Ω	0.03Ω
L_c	$0.35mH$	$0.35mH$		$0.35mH$	$0.35mH$
k_{Pv}	0.1	0.1		0.05	0.05
k_{Iv}	420	420		390	390
k_{Pc}	15	15		10.5	10.5
k_{Ic}	20000	20000		16000	16000
R_{l1}	0.23Ω	R_{l2}	0.35Ω	R_{l3}	0.23Ω
L_{l1}	$318\mu H$	L_{l2}	$1847\mu H$	L_{l3}	$318\mu H$
Load1	140Kw, 20Kvar(0-4s)			140Kw, 20Kvar(4-6s)	
Load2	100Kw, 30Kvar(0-4s)			150Kw, 50Kvar(4-6s)	

Table 2: Cost Coefficients of DGs

DG i	DG 1	DG 2	DG 3	DG 4
$\gamma_i (\$/Kw^2h)$	0.096	0.072	1.105	0.082
$\rho_i (\$/Kwh)$	1.22	3.41	2.53	4.02
$\delta_i (\$/h)$	51	31	78	42

To show the dynamic behavior of the proposed frequency controller, this simulation is divided into three stages: Stage1(0-2s): only primary droop control is used; Stage2(2-6s): the proposed secondary frequency controller is applied; Stage3(4-6s): the load 2 is changed at 4s. The settling time of secondary frequency control is $T_1 \leq 3.11s$ based on the selected parameters. Fig.4-Fig.6 show the output frequency, output active power, and incremental cost, respectively.

As Fig. 4 shows, the DGs output frequency ω_i is restored to the nominal value $\omega^* = 50\text{Hz}$ in fixed-time T_1 for both controller activation and load changing scenarios. Fig. 6 shows that all DG's incremental costs are equal in fixed-time T_1 , which means that the optimal active power sharing is achieved in fixed time.

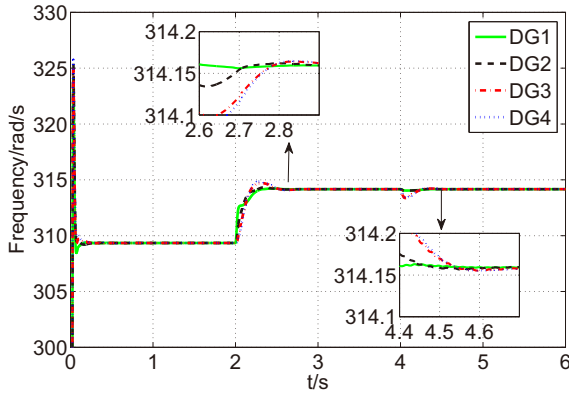


Fig. 4: The output frequency

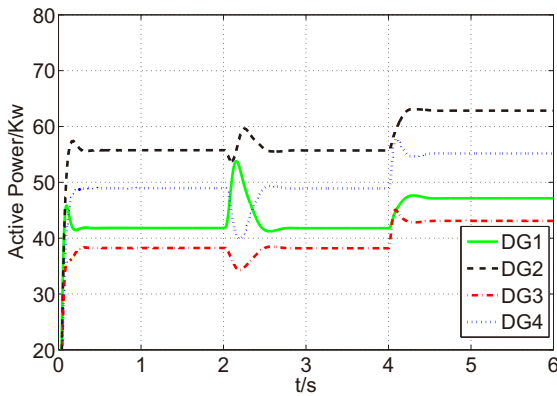


Fig. 5: The output active power

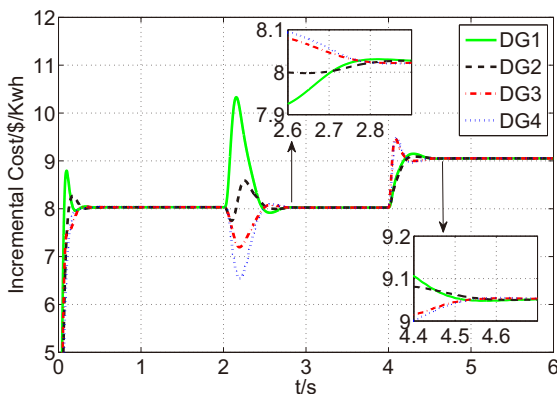


Fig. 6: The incremental cost

5 Conclusion

This problem of frequency restoration and optimal active power sharing in islanded microgrids was solved in this paper. Based on the incremental cost droop approach and

the distributed fixed-time approach, the distributed frequency controller was designed and analyzed. Simulation results verified the effectiveness of the proposed frequency controller. The further work will further consider the inequality constraints of generation capacity.

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